

# Day 5: Amazon S3 & Identity & Access Management (IAM)

*Object Storage, Identity Management, Policies, and Least Privilege Principle*

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## Module 1: Amazon Simple Storage Service (S3)

Amazon S3 is a highly durable, scalable object storage platform. S3 stores files as 'objects' inside 'buckets' rather than blocks or hierarchical file trees. It is built to achieve 99.999999999% durability (11 9s) by duplicating files across three distinct Availability Zones within an AWS Region.

### S3 Storage Classes

| Class                   | Durability | Access Retrieval            | Optimal Use Case                                               |
|-------------------------|------------|-----------------------------|----------------------------------------------------------------|
| S3 Standard             | 11 9s      | Immediate<br>(milliseconds) | Frequently accessed files, active web pages, dynamic assets    |
| S3 Standard-IA          | 11 9s      | Immediate<br>(milliseconds) | Infrequently accessed files that need fast retrieval (backups) |
| S3 Intelligent-Tiering  | 11 9s      | Immediate<br>(milliseconds) | Data sets with unknown, unpredictable access trends            |
| S3 Glacier Flexible     | 11 9s      | 1 minute to 5 hours         | Archived files, compliance records, historical files           |
| S3 Glacier Deep Archive | 11 9s      | 12 hours                    | Cold storage datasets, logs kept for years rarely retrieved    |

## Module 2: AWS IAM & Access Security

AWS Identity and Access Management (IAM) controls authentication and authorization. It determines which identities (Users, Groups, Roles) have access to specific AWS APIs and resources.

### Principle of Least Privilege

The Principle of Least Privilege (PoLP) states that system accounts and users should be granted only the minimum access required to execute their specific job duties. Limiting permission scopes prevents accidental deletions, credential leakage damage, and unauthorized configuration modifications.

### Types of IAM Policies

- **AWS Managed Policies:** Standard permissions created and maintained by AWS (e.g. AdministratorAccess). Available to all AWS customers but cannot be modified.
- **Customer Managed Policies:** Standalone policies created by the user for precise account control. Recommended for production environments.
- **Inline Policies:** Policies embedded directly into a single IAM user, group, or role. They are destroyed if the parent identity is deleted.

- **Resource-Based Policies:** Policies attached directly to resources rather than users (e.g., S3 Bucket Policies).